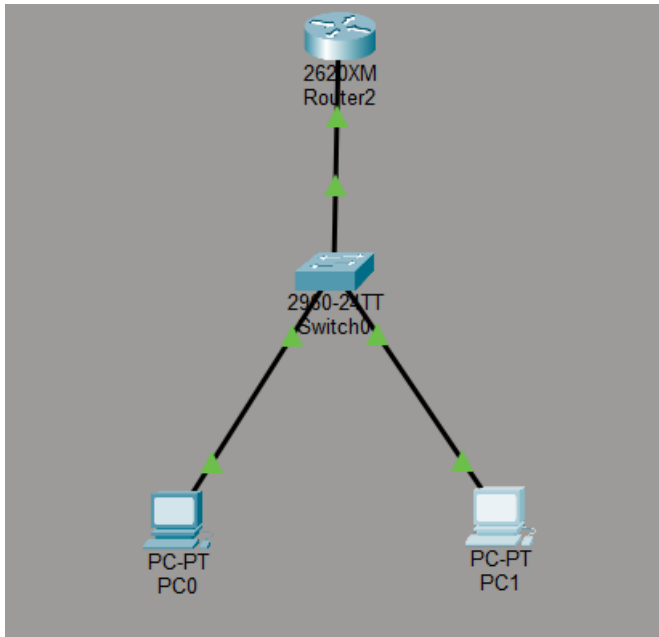


Q-Configure a DHCP server in a network simulator (such as Cisco Packet Tracer or GNS3) and intentionally set an incorrect default gateway; connect a client device and verify the assigned gateway, then describe how you would identify and fix this issue.

To configure a router as a DHCP server, intentionally assign an incorrect default gateway to clients, verify the assigned gateway, identify the problem, and correct the configuration.

TOPOLOGY:



Router Configuration

```
enable
configure terminal

interface fa0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit

ip dhcp excluded-address 192.168.1.1

ip dhcp pool LAN
network 192.168.1.0 255.255.255.0
default-router 192.168.1.254
dns-server 8.8.8.8
end
```

PC Configuration

1. Open **Desktop** → **IP Configuration**.
2. Select **DHCP**.
3. The PCs receive IP addresses automatically.

Example:

Device	IP Address	Default Gateway
PC0	192.168.1.2	192.168.1.254
PC1	192.168.1.3	192.168.1.254

Verification

Run the following command on **PC0**:

ipconfig

Output:

IP Address.....192.168.1.2
Subnet Mask.....255.255.255.0
Default Gateway.....192.168.1.254

The default gateway is incorrect because the router's actual IP address is **192.168.1.1**.

Problem Identification

The issue was identified by:

- Checking the client's IP configuration using ipconfig.
- Comparing the assigned default gateway with the router interface IP address.
- Reviewing the DHCP pool configuration on the router.

The DHCP pool contained:

default-router 192.168.1.254

which is an incorrect gateway.

Solution

Correct the DHCP configuration:

configure terminal

```
ip dhcp pool LAN  
no default-router 192.168.1.254  
default-router 192.168.1.1
```

end

Renew the DHCP lease on both PCs by selecting **DHCP** again in the IP Configuration window.

Run:

```
ipconfig
```

output:

```
IP Address.....192.168.1.2  
Subnet Mask.....255.255.255.0  
Default Gateway.....192.168.1.1
```

The DHCP server initially assigned an incorrect default gateway (192.168.1.254) to the client PCs. The issue was identified using the ipconfig command and by reviewing the router's DHCP configuration. After changing the default-router option to the correct router IP (192.168.1.1) and renewing the DHCP lease, both PCs received the correct default gateway and the configuration was successfully corrected.